# Syed Usman

Senior DevOps & Platform Engineer, Berlin
Permanent residence, no visa sponsorship needed

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## Summary


Engineer with a decade across hardware, cloud and Kubernetes: from IoT products monitoring 900+ lifts, through two years as tech lead of 8 engineers, to a Kubernetes learning platform for 10,000+ learners. For the past year almost all of my infrastructure work has gone through AI agents (Claude Code with skills, a team-built harness and the BMAD Method), roughly quadrupling my tickets and story points per sprint, and I design the guardrails that keep them safe on real infrastructure. Open to staff, lead and engineering manager roles, and to freelance projects.


Throughout my career, I have gained experience in the following areas:

- **AI:** Claude Code (skills, subagents, hooks), a team-built harness, the BMAD Method, least-privilege agent identity, sandboxed agent execution
- **Cloud Infrastructure:** AWS, Azure, Hetzner, Scaleway, IONOS
- **IaC & Automation:** Terraform, Ansible, Helm, GitOps, Renovate
- **Kubernetes & Containerization:** Kubernetes, Docker, Velero, Keycloak
- **CI/CD:** GitHub Actions, GitLab CI, Atlantis
- **Observability & SRE:** Prometheus, Grafana, New Relic, OpenTelemetry, Datadog, load testing, backup & recovery, incident response & on-call
- **Backend:** Python, PostgreSQL, Redis, Kafka

**Languages:** English (native), German (B1, improving). **Work setup:** own office near Berlin, in person for onboarding, workshops and key moments.

## Experience

- **Senior DevOps/Infrastructure Engineer at Multiverse** Aug. 2024 – Present, Berlin
  *Joined StackFuel, acquired by Multiverse in 2025*
  - **Load Testing & Scaling to 200 Concurrent Users** Diagnosed why a JupyterHub load test on Kubernetes stalled at ~170 of 200 users with 15–23% failed logins; over ten instrumented runs, traced it to node autoscaling reconfiguring the load balancer mid-test. Fixed in Terraform and Helm: 200/200 users, zero failures. Write-up.
  - **Platform Migrations & Cost** Moved Coding Labs (JupyterHub) and the analytics stack (Airflow, Metabase) from manual and on-premise servers to autoscaling Kubernetes on Hetzner and Scaleway for 10,000+ learners, cutting infrastructure cost by 30%+. Upgraded clusters across three environments to Kubernetes 1.35; manage Keycloak realms and DNS in Terraform.
  - **Backup & Recovery** Built Kubernetes volume backups with Velero and Kopia (90-day retention), a restore runbook from single file to full environment, and a script that computes the minimal restore chain across volumes. Drills on the two main databases measured a restore time of 1–4 hours. Write-up.
  - **AI Agent Identity, Access & Sandboxing** Designed a dedicated, least-privilege identity for Claude: short-lived OIDC-federated credentials, network allow-listing, and full audit logging, scoped read-only. Ran it unattended in Docker sandboxes to investigate and resolve infrastructure issues, isolated from production. Write-up.
  - **AI-Assisted Delivery Pipeline** Co-built a Claude Code pipeline taking Jira tickets to merge requests with competing planners, review subagents and two human approval gates; designed its default-deny tool-access hook and treated ticket text as untrusted input. Deployed a company-wide AI experimentation platform with Helm.
  - **Skills for Recurring Work** Turned recurring work into Claude Code skills, including skills that learn colleagues' code-review patterns and check merge requests for them before review.
  - **Sandboxed Automation Platform** Deployed n8n with Ansible and Keycloak SSO; isolated its AI assistant in sysbox-runc sandboxes with no privileged fallback.
  - **Monitoring & IaC Modernization** Standardized New Relic monitoring (agents via Ansible, alerts via Terraform); split a monolithic Ansible codebase into modular roles with GitLab CI; rolled out Renovate across repositories.

- **Senior Cloud Infrastructure Engineer at Jillion** Nov. 2023 – June 2024, Berlin

  - **Terraform and GitOps** Designed and deployed Staging and Production environments on AWS with Terraform and GitOps, including IAM permissions as code.
  - **CI/CD and Automation** Built GitHub Actions pipelines for infrastructure deployment, SonarQube quality checks, tests and image builds to ECR, with runners on Kubernetes.
  - **Kubernetes and Optimization** Deployed ArgoCD to manage deployments on AWS EKS. Improved cluster resilience with capacity reservations, pod priority classes, eviction policies, and resource requests/limits enforcement via admission control.
  - **Observability and Self-Service Platform** Deployed OpenTelemetry and Datadog for service instrumentation, metrics, and log aggregation. Built self-service jobs enabling engineers to provision infrastructure, manage secrets, and deploy K8s workloads independently.
  - **Deployment, VPN and Access Control** Developed Helm charts for internal and external services (Gloo, Oathkeeper, OpenTelemetry, Datadog). Configured Pritunl site-to-site VPN with declarative access control for developers and infrastructure teams.

- **Freelance DevOps Engineer** Jun. 2023 – Nov. 2023, Berlin

  *Through a contracting agency*

  - **E-commerce Expansion** Built the infrastructure for a US dental supplies retailer's high-volume Magento Cloud store, with many connected platforms, to expand into a new country.

- **Munich Re (relayr GmbH)** Jul. 2018 – June 2023, Berlin

  *relayr: Munich Re's industrial IoT company*
  *Solutions Architect (Tech Lead)* *May 2021 – June 2023*

  - **Team Leadership** Tech lead of a cross-functional team of 8 engineers: hiring and technical interviews, 1:1s and mentoring, sprint and roadmap planning, requirement workshops with clients, and incident response with postmortems and on-call.
  - **Cloud & Kubernetes Strategy** Led company-wide adoption of Kubernetes and GitOps across AWS and Azure for 5+ client platforms, using Terraform with PR automation via Atlantis to streamline infrastructure changes as a team.
  - **Equipment as a Service (EaaS) – Industrial machine rental platform** Built and ran Kubernetes on-premise and on EKS, with GitHub CI/CD, large Azure Postgres and Prometheus/Grafana monitoring.
  - **Cincinnati – Hydraulic Press & Laser Cutting Machine Monitoring** Architected a near-real-time IoT backend on Kubernetes and Azure, moving a Django monolith to Flask microservices with Prometheus and EFK.
  - **Gropyus & Ecospray – Smart Building and Maritime IoT Platforms** Built cloud infrastructure and observability (Telegraf, InfluxDB, Grafana, Prometheus) for IoT monitoring platforms on AWS, contributing to a ~12% reduction in vessel fuel costs for Ecospray and streamlined field onboarding for Gropyus.

  *Senior Solutions Engineer* *May 2019 – Apr. 2021*

  - **Franz – IoT Elevator Monitoring Platform** Took Franz to a working MVP in under 6 months with a cross-functional team (FE, BE, firmware, Linux); the platform now monitors 900+ lifts in production. Built the edge software layer (Python, MQTT, Buildroot) and an OTA update system for embedded Linux devices.

  *Solutions Engineer* *Jul. 2018 – Apr. 2019*

  - **Lift Monitoring Backend** Built AWS infrastructure for lift monitoring: dockerized microservices on Kubernetes, a Python gateway agent, Terraform across environments, Prometheus/Grafana alerting and Jenkins pipelines; worked with vendors to industrialize gateways, sensors and firmware.

- **Software Engineer at Colormass GmbH** Jul. 2017 – Apr. 2018, Berlin

  - **3D Scene Editor & Draco Port** Built a three.js scene editor where clients set up scenes and camera angles for renders made in 3ds Max; ported Google's Draco geometry compression from C++ to JavaScript to compress and decompress heavy models in the browser.
  - **Infrastructure Migration & Automation** Moved a Django monolith from App Engine to Kubernetes microservices with Jenkins CI/CD; built a GCP render farm in Python that halved render times.

- **Software Engineer at YotaScale Inc.** Aug. 2015 – Jun. 2017, Palo Alto & Lahore

  - **Data Processing and Cost Analysis** Built an AWS cost/utilization data pipeline (Kinesis, Lambda, Redshift) with TypeScript and Flask dashboards to surface waste; automated deployments with Jenkins and migrated on-prem Postgres to AWS RDS.

## Education

- **National University of Sciences & Technology (NUST)** — Islamabad
  *Bachelors in Software Engineering; Gold Medallist* — *2011 – 2015*
- **AWS Certified** — Islamabad
  *Solutions Architect – Associate* — *2018*